

Policy as Strategic Investment: A Recursive Monetary–Fiscal Game Calibrated to Australia

Monetary–Fiscal Coordination Project

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Abstract

This paper turns a two-stage monetary–fiscal policy game into a finite- and infinite-horizon recursive game. Instrument-adjustment costs are fixed primitives. Current monetary and fiscal choices instead constitute strategic investments because they become payoff-relevant state variables and change the other authority’s later feedback action. We solve the exact two-stage game by backward induction, arbitrary finite horizons using coupled Riccati recursions, a stationary Markov-perfect equilibrium, an open-loop Nash benchmark on the identical economic system, and cooperation under a fixed social objective. The model is calibrated using quarterly Australian transmission estimates. The strategic channel is present: in the terminal subgame, $\partial f_2/\partial m_1 = 0.0086$ and $\partial m_2/\partial f_1 = 0.0024$. It is nevertheless small at the baseline calibration. Forty-quarter feedback Nash social losses are 0.2037 after a demand contraction and 0.7078 after a cost-push shock, compared with open-loop losses of 0.2037 and 0.7078. The conclusion is substantive rather than computational: adjustment costs make policy durable, but a large strategic-investment distortion additionally requires strong cross-authority transmission and a material conflict over the induced response. The framework makes those objects measurable and separates them from ordinary persistence and own-policy smoothing.

Keywords: monetary–fiscal coordination; strategic investment; Markov-perfect equilibrium; adjustment costs; policy games; Australia.

JEL codes: E52, E58, E61, E62, C73.

1 Introduction

Monetary and fiscal policies affect the same output, inflation and debt outcomes but need not be chosen by an authority with a common objective. The resulting interaction is not exhausted by a static Nash equilibrium. A policy choice made today can change the position inherited by tomorrow’s policymakers and thereby alter their incentives. Current policy is then an investment in a future strategic position.

The distinction matters for the interpretation of adjustment costs. A term such as $\lambda_M(m_t - m_{t-1})^2$ does not merely smooth the interest rate. It makes m_t part of tomorrow’s state. If fiscal policy responds to that state, the monetary authority may move m_t partly to influence future fiscal policy. The corresponding fiscal adjustment term creates the reciprocal channel. The adjustment coefficients λ_M and λ_F are technological or institutional primitives in this paper; the authorities do not choose their own objective coefficients.

This formulation follows the logic of strategic investment in industrial organisation. In [Dixit \(1980\)](#) and [Fudenberg and Tirole \(1983\)](#), a durable first-stage choice commits a player to a position that changes a rival’s continuation behaviour. [Fudenberg and Tirole \(1984\)](#) and [Bulow, Geanakoplos and Klemperer \(1985\)](#) show that the sign of the distortion depends on two distinct objects:

whether the investment makes the player tough or soft, and whether continuation actions are strategic complements or substitutes. Adjustment costs provide durability in the present model, but they do not by themselves determine either object.

Dynamic policy games apply closely related logic. [Oudiz and Sachs \(1984\)](#) distinguish precommitment, time consistency, feedback and cooperation in macroeconomic coordination. [Tabellini \(1986\)](#) studies monetary policy, deficits and public debt in a dynamic linear-quadratic game. Most directly, [Beetsma and Bovenberg \(1997\)](#) treats public debt as a strategic fiscal state that affects later monetary behaviour. [Gnocchi \(2013\)](#) demonstrates that monetary-fiscal outcomes depend critically on the strategies to which the central bank can commit. These contributions imply that strategy space and state definition are economic assumptions, not computational details.

The paper makes four contributions. First, it supplies the exact backward-induction extension of the original two-choice game. Second, it decomposes the stage-one first-order condition and identifies the rival-response term, rather than labelling every continuation effect strategic. Third, it compares feedback and open-loop Nash equilibria on identical equations, shocks, weights and adjustment costs. Fourth, it documents convergence to a stationary feedback equilibrium and conducts a multi-start numerical equilibrium search.

2 The recursive Australian policy model

2.1 Economic block

Let positive m_t denote monetary tightening and positive f_t denote a fiscal expansion. The empirical recursive block is

$$x_t = \rho_x x_{t-1} + \sigma \pi_{t-1} - \sigma m_t + \psi f_t + d_t, \quad (1)$$

$$\pi_t = \rho_\pi \pi_{t-1} + \kappa x_t + u_t, \quad (2)$$

$$b_{t+1} = \rho_b b_t + \chi_m m_t + \chi_f f_t - \chi_\pi \pi_t, \quad (3)$$

$$d_{t+1} = \rho_d d_t + \varepsilon_{t+1}^d, \quad u_{t+1} = \rho_u u_t + \varepsilon_{t+1}^u. \quad (4)$$

The policy state before decisions at t is

$$s_t = (x_{t-1}, \pi_{t-1}, b_t, m_{t-1}, f_{t-1}, d_t, u_t)'. \quad (5)$$

Equations (1)–(4) can be collected as

$$s_{t+1} = A s_t + B_M m_t + B_F f_t + \varepsilon_{t+1}. \quad (6)$$

The unit entries in B_M and B_F place the current instruments in the next period's lagged-instrument states. They are the formal link between the two-stage game and its dynamic extension.

The block is deliberately transparent. The output and inflation equations are hybrid, backward-looking empirical relations rather than the fully forward-looking New Keynesian IS and Phillips curves. This permits an exact recursive solution and a clean comparison of equilibrium concepts. It also sets a boundary on interpretation: the exercise does not yet incorporate the additional commitment channel operating through private expectations.

2.2 Agency mandates and true social welfare

Authority $j \in \{M, F\}$ minimizes

$$L_j = E_0 \sum_{t=0}^{T-1} \beta_j^t [\omega_\pi^j \pi_t^2 + \omega_x^j x_t^2 + \omega_b^j b_{t+1}^2 + \omega_m^j m_t^2 + \omega_f^j f_t^2 + \lambda_M^j (m_t - m_{t-1})^2 + \lambda_F^j (f_t - f_{t-1})^2]. \quad (7)$$

The central bank places greater relative weight on inflation; the fiscal authority places greater relative weight on output. These weights are held fixed across every solution concept.

Cooperation minimizes a third loss L_S of the same form. It represents true social welfare and is not the sum of the two institutional mandates. This separation prevents the definition of cooperation from changing whenever the agencies' delegated objectives change.

In matrix form, the stage loss is

$$\ell_j(s_t, m_t, f_t) = \begin{bmatrix} s_t' & m_t & f_t \end{bmatrix} Q_j \begin{bmatrix} s_t' & m_t & f_t \end{bmatrix}'. \quad (8)$$

3 Strategic investment and equilibrium concepts

3.1 The exact two-stage game

At stage 2, the two authorities play a simultaneous Nash game conditional on s_2 . Its linear decision rules are

$$m_2 = K_{M,2}s_2, \quad f_2 = K_{F,2}s_2. \quad (9)$$

At stage 1, each authority understands that its action changes s_2 and hence both rules in (9). For the monetary authority,

$$0 = \ell_{m_1}^M + \beta_M B'_M \nabla V_2^M(s_2). \quad (10)$$

Using the terminal monetary first-order condition gives the envelope decomposition

$$\nabla V_2^M = \underbrace{\ell_{s_2}^M}_{\text{direct state effect}} + \underbrace{K'_{F,2} \ell_{f_2}^M}_{\text{rival-response effect}}. \quad (11)$$

The term $K'_{M,2} \ell_{m_2}^M$ vanishes by the monetary authority's own terminal first-order condition. Therefore

$$\mathcal{S}_M = \beta_M B'_M K'_{F,2} \ell_{f_2}^M \quad (12)$$

is the monetary strategic-investment wedge. Similarly,

$$\mathcal{S}_F = \beta_F B'_F K'_{M,2} \ell_{m_2}^F. \quad (13)$$

This isolates the strategic term from own adjustment costs, ordinary macro persistence and debt dynamics. Strategic investment exists only if the current policy changes the rival's later action and that action matters for the current policymaker's loss.

3.2 Finite-horizon feedback Nash

Suppose the continuation values are quadratic, $V_{t+1}^j(s) = s'P_{t+1}^j s$. At each date the simultaneous policy first-order conditions are

$$\begin{bmatrix} Q_{mm}^M + \beta_M B_M' P_{t+1}^M B_M & Q_{mf}^M + \beta_M B_M' P_{t+1}^M B_F \\ Q_{fm}^F + \beta_F B_F' P_{t+1}^F B_M & Q_{ff}^F + \beta_F B_F' P_{t+1}^F B_F \end{bmatrix} \begin{bmatrix} m_t \\ f_t \end{bmatrix} = - \begin{bmatrix} Q_{ms}^M + \beta_M B_M' P_{t+1}^M A \\ Q_{fs}^F + \beta_F B_F' P_{t+1}^F A \end{bmatrix} s_t. \quad (14)$$

Solving (14) yields $K_{M,t}$ and $K_{F,t}$. With $A_t^c = A + B_M K_{M,t} + B_F K_{F,t}$, each value matrix is updated by

$$P_t^j = S_t' Q_j S_t + \beta_j (A_t^c)' P_{t+1}^j A_t^c, \quad S_t = \begin{bmatrix} I \\ K_{M,t} \\ K_{F,t} \end{bmatrix}. \quad (15)$$

Starting from $P_{T+1}^j = 0$ and working backwards produces a subgame-perfect feedback Nash equilibrium. For $T = 2$, this is exactly the original two-stage logic. As T grows, the initial rules converge to a stationary coupled- Riccati fixed point.

3.3 Open loop, cooperation and equilibrium selection

In the open-loop Nash benchmark, both authorities select complete paths at date zero. Later controls therefore do not respond to a deviation that changes an inherited state. We solve this game by stacking (6) and the quadratic losses, not by changing the economy. The difference between feedback and open loop consequently identifies the value of state-contingent continuation behaviour in this calibration.

The cooperative regulator uses the same backward recursion but chooses both instruments to minimize L_S . Finally, the stationary Nash solution is computed from several dispersed initial value matrices. Finding the same stabilising fixed point from these starts is useful evidence, but is not a mathematical proof that no other Markov-perfect equilibrium exists.

4 Australian calibration

The quarterly output and inflation equations use the 1993Q1–2019Q4 Australian estimates employed in the companion model. The cached source data come from the Reserve Bank of Australia's statistical tables. The debt block and loss weights are transparent calibrations rather than structural estimates. Table 1 reports the principal values.

Table 1: Baseline calibration

Parameter	Value	Interpretation
ρ_x	0.62875	output-gap persistence
ρ_π	0.56255	inflation persistence
σ	0.03695	monetary transmission to output
ψ	0.05142	fiscal transmission to output
κ	0.15511	Phillips slope
ρ_b	0.995	debt persistence
$(\chi_m, \chi_f, \chi_\pi)$	(0.015, 0.060, 0.100)	debt transmission
(λ_M, λ_F)	(0.050, 0.040)	primitive adjustment weights
$(\omega_\pi^M, \omega_x^M)$	(1.50, 0.25)	monetary mandate
$(\omega_\pi^F, \omega_x^F)$	(0.50, 0.75)	fiscal mandate
$(\omega_\pi^S, \omega_x^S)$	(1.00, 0.50)	social objective

The experiments use a one-standard-deviation demand contraction of 0.464 and a cost-push innovation of 0.670. Innovations after the initial impulse are zero in the plotted paths. Under linear-quadratic certainty equivalence, additive shock variances affect expected loss but not the policy gains. Persistent shocks could be included by changing ρ_d and ρ_u without altering the solution method.

5 Results

5.1 The two-stage mechanism is present but small

Table 2 reports the exact stage-2 response coefficients and the realised strategic component of the stage-1 half-gradient. The two cross responses are positive under the sign convention used here. A monetary tightening today raises the fiscal expansion chosen in the continuation game; a fiscal expansion today raises subsequent monetary tightening. The policy positions are therefore locally complementary in this cross-stage sense.

Table 2: Two-stage strategic-investment diagnostics

Diagnostic	Demand contraction	Cost-push shock
$\partial f_2 / \partial m_1$	0.00856	0.00856
$\partial m_2 / \partial f_1$	0.00237	0.00237
Monetary strategic wedge $\mathcal{S}_M$	0.000055	0.000021
Fiscal strategic wedge $\mathcal{S}_F$	0.000011	0.000004
Monetary total continuation derivative	0.002173	−0.003395
Fiscal total continuation derivative	−0.005328	0.001140

The signs establish the mechanism; their scale disciplines its interpretation. For the demand shock, the monetary strategic term is about 2.6 per cent of the absolute monetary continuation derivative. The fiscal share is roughly 0.2 per cent. The remaining continuation incentive comes from direct macro persistence, debt and own-policy smoothing. Thus it would be incorrect to call the entire continuation term strategic investment.

Figure 1 compares feedback and open-loop policies in the exact two-stage experiment. Their difference is visible numerically but small. This is not a failure of backward induction: it is the implication of weak estimated instrument transmission and small cross-stage response slopes.

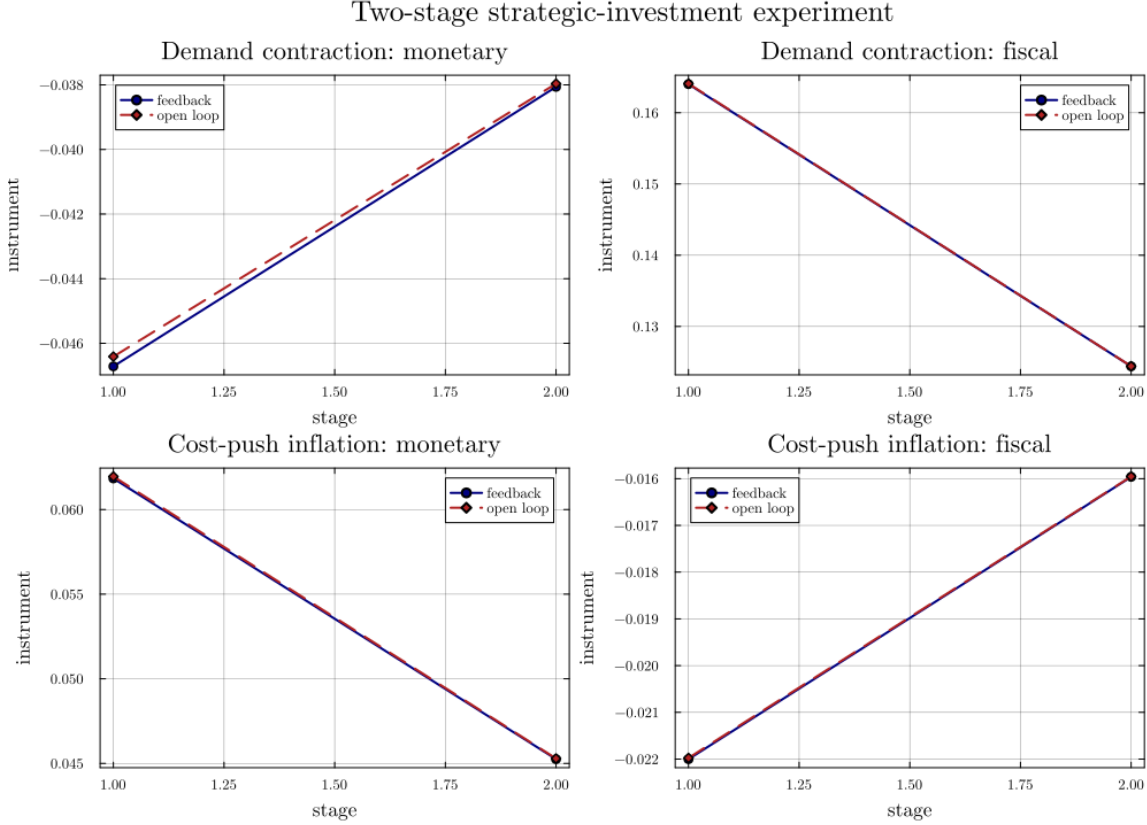


Figure 1: Exact two-stage feedback and complete-path open-loop policies.

5.2 The dynamic solution

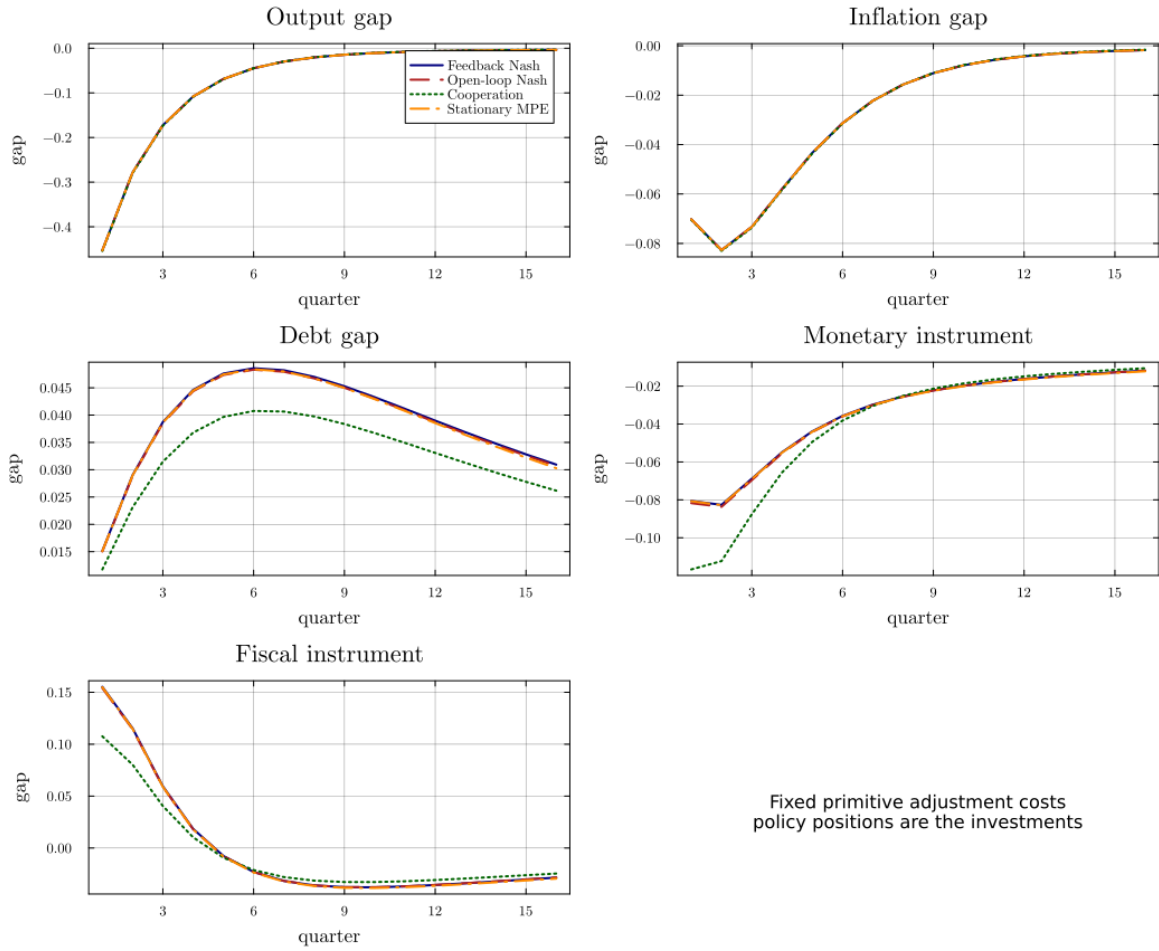
Table 3 reports forty-quarter losses. Feedback Nash and open loop are close under both shocks. Cooperation matters more, especially for the allocation of instruments, but even its welfare gain is modest at this local calibration.

Table 3: Forty-quarter social loss

Regime	Demand contraction	Cost-push shock
Feedback Nash	0.2037	0.7078
Open-loop Nash	0.2037	0.7078
Cooperation	0.2027	0.7074
Stationary MPE rule	0.2037	0.7079

Figures 2 and 3 show the corresponding impulse responses. The stationary MPE and the forty-period feedback solution nearly coincide at the beginning of the sample. Open loop is also close, consistent with the small strategic wedges. Cooperation uses a different policy mix: following a demand contraction it relies relatively more on monetary easing and less on fiscal expansion.

One-s.d. Australian demand contraction



Fixed primitive adjustment costs
policy positions are the investments

Figure 2: Dynamic responses to a one-standard-deviation Australian demand contraction.

One-s.d. Australian cost-push shock

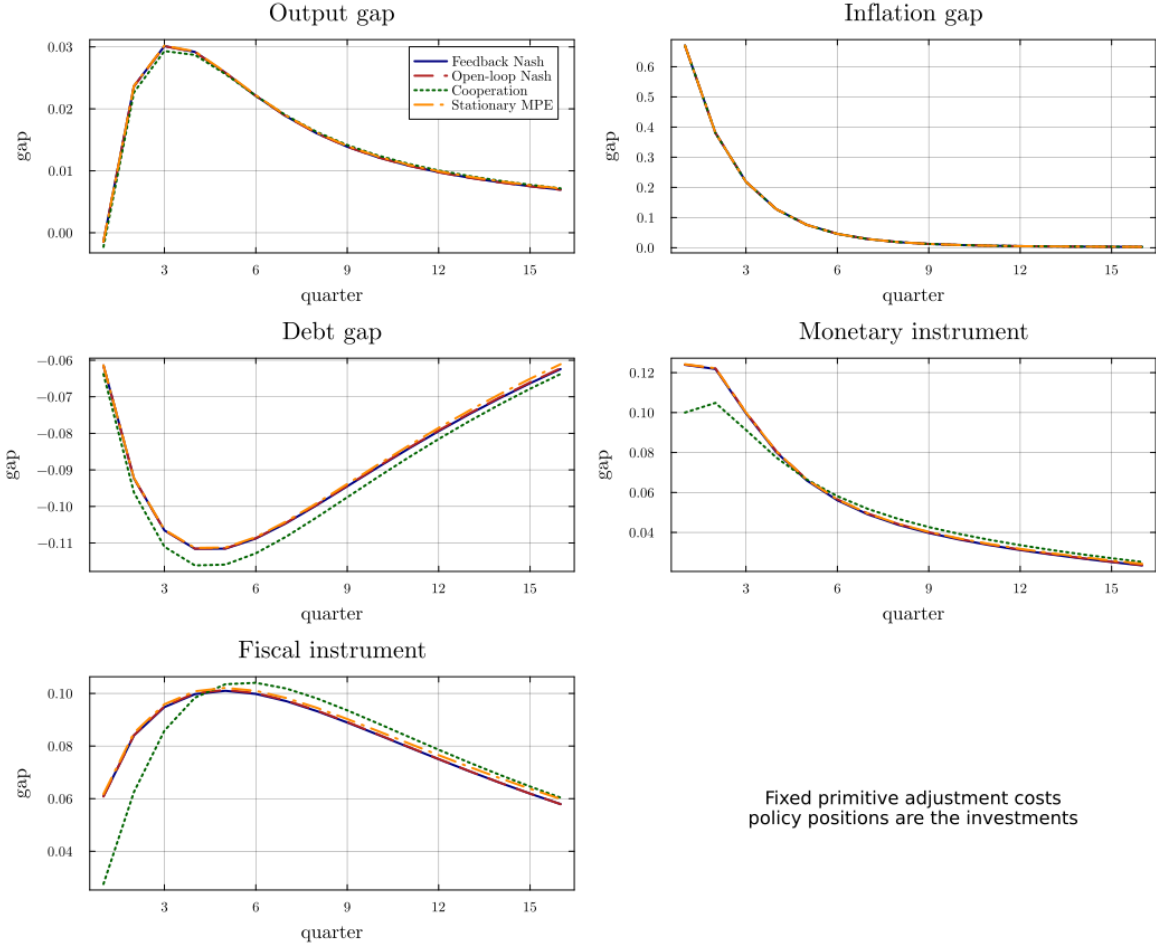


Figure 3: Dynamic responses to a one-standard-deviation Australian cost-push shock.

5.3 Horizon convergence and stationary equilibrium

The finite-horizon rule converges steadily toward the stationary feedback rule. At horizon 40 its Euclidean coefficient distance is 0.00534; at horizon 80 it is 2.1×10^{-5} . The stationary closed-loop spectral radius is 0.9383, so the selected equilibrium is locally stable.

Six Riccati initialisations ranging from zero to one hundred times the stage- loss state matrix all converged to the same rule to numerical tolerance. The search therefore found 1 distinct stable solution. This is reassuring for the baseline but should not be read as a global uniqueness theorem. If stronger complementarity creates multiple fixed points, they should be sought using continuation methods and direct solution of the coupled algebraic equations.

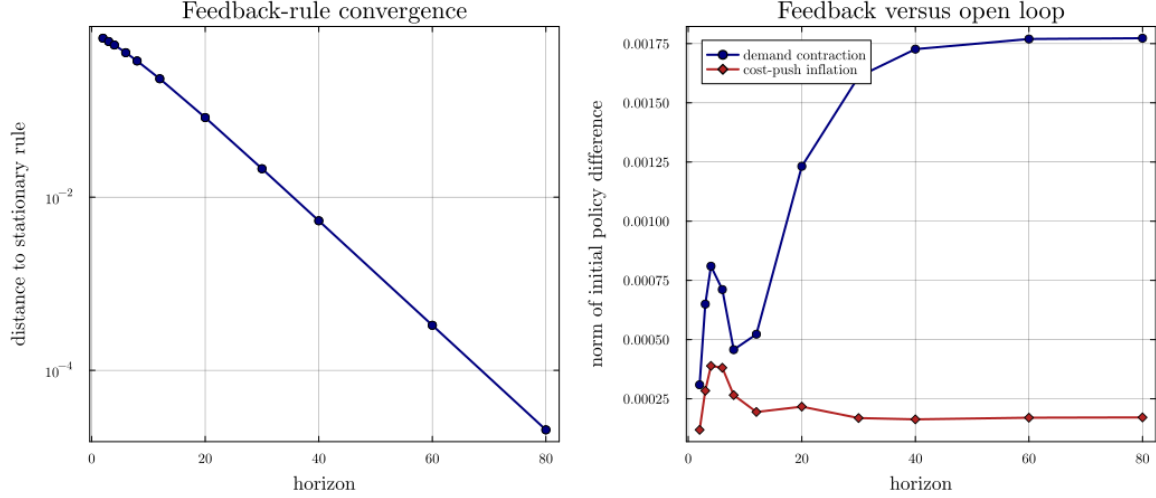


Figure 4: Convergence to the stationary feedback rule and the feedback–open-loop gap.

5.4 Why larger adjustment costs do not automatically create a large game

Figure 5 scales both primitive adjustment-cost weights while holding structural coefficients and mandate weights fixed. The cross-stage responses initially increase slightly and then decline. Very high adjustment costs make the policy position durable, but also suppress the rival’s ability to react. Accordingly, durability and strategic responsiveness work in opposite directions.

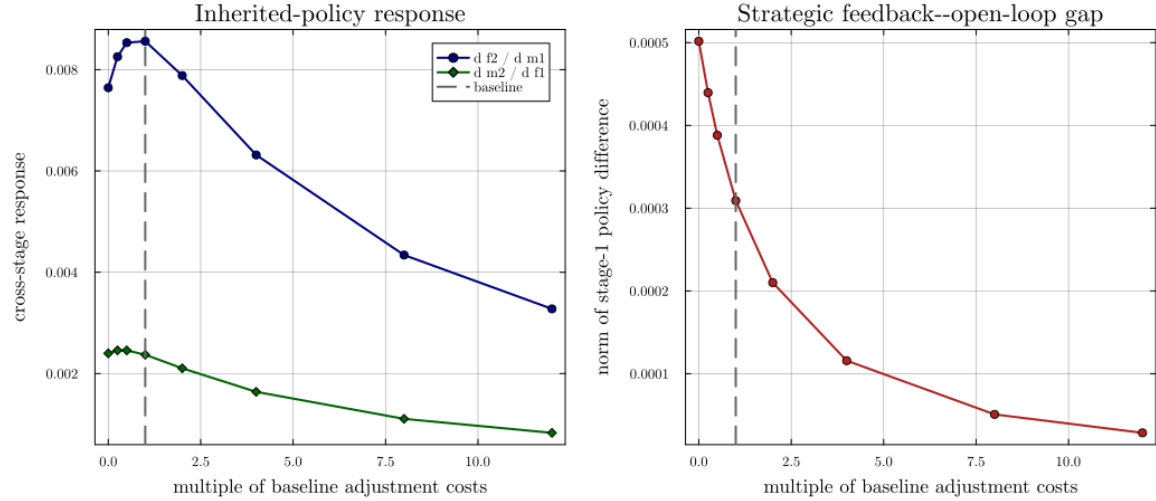


Figure 5: Two-stage strategic investment as primitive adjustment costs vary.

This result clarifies the theoretical mechanism. Adjustment costs make policy a state, which is necessary for the particular lagged-instrument channel. They do not ensure a large strategic motive. The motive also requires

1. a sizeable effect of current policy on a state observed in the next continuation game;
2. a sizeable response of the rival’s policy to that state; and
3. a sizeable marginal effect of the rival’s response on the investing authority’s objective.

The Australian baseline is weak primarily on the second and third margins. The experiment therefore points toward transmission, mandate divergence, debt exposure and shock persistence as the economically relevant sensitivity axes.

6 Relation to fiscal dominance

The model contains a strategic-debt channel but not a full fiscal-theory price- level mechanism. Fiscal expansion raises debt; debt is persistent; and both future policy rules respond to debt. Fiscal policy can therefore select a future state that induces monetary accommodation or tightening. This is close to the strategic debt management mechanism in [Beetsma and Bovenberg \(1997\)](#) and to the dynamic conflict studied by [Tabellini \(1986\)](#).

Fiscal dominance in the stronger sense concerns the inability or unwillingness of monetary policy to stabilise prices without fiscal backing. Nesting that literature fully requires nominal government liabilities, a consolidated intertemporal budget constraint and an explicit active/passive policy classification. The present model should therefore be interpreted as capturing one strategic route toward fiscal pressure, rather than as a complete fiscal theory of the price level.

The strategy-space result of [Gnocchi \(2013\)](#) is especially relevant. An unrestricted monetary commitment can alter fiscal incentives in ways a simple interest-rate path cannot. In the present language, an open-loop path, a Markov feedback rule and an institutionally committed rule are different objects. The game over adjustment coefficients belongs to the last category and should be studied separately from policy-state investment.

7 Steady state, shocks and coordination

With zero gaps and no shocks, $s = 0$ and $(m, f) = (0, 0)$ form the common steady state. The realised strategic-investment wedge is then zero because every quadratic marginal loss is zero. This does not eliminate the game: the policy derivatives in [Table 2](#) remain features of the continuation rules. A shock moves the system away from the steady state and activates the latent interaction.

Strategic complementarity should also be distinguished from equilibrium multiplicity. Positive cross responses can amplify policies while the joint best-response system remains a contraction and the equilibrium remains unique. Coordination on the “right” equilibrium becomes a separate issue when the feedback mapping ceases to be contractive or the coupled Riccati system has multiple stabilising solutions. The baseline exhibits complementarity but not numerical evidence of multiplicity.

8 Limitations and the next structural extension

Three limitations are important. First, the empirical block is reduced form; the private sector does not solve forward-looking Euler and pricing equations. A structural extension should solve

$$\begin{aligned} x_t &= E_t x_{t+1} - \sigma(m_t - E_t \pi_{t+1}) + \psi f_t + d_t, \\ \pi_t &= \beta E_t \pi_{t+1} + \kappa x_t + u_t \end{aligned}$$

jointly with the Markov policy rules. Anticipated future policy would then affect current activity and inflation, creating an additional commitment channel.

Second, the local quadratic model cannot generate occasionally binding fiscal limits, the effective lower bound or nonlinear debt-risk premia. Those features could enlarge strategic complementarities and create multiple equilibria. Third, the Australian exercise calibrates the debt block and

objectives. Estimating the policy reaction and debt-transmission parameters would be necessary before attaching empirical confidence intervals to the strategic wedges.

9 Conclusion

The dynamic version of the two-stage game is a feedback Nash game in which today’s monetary and fiscal choices become tomorrow’s state. Backward induction, not a separate game over objective coefficients, is the appropriate extension. The exact two-stage decomposition verifies a strategic-investment motive in the Australian calibration. Its magnitude is small, and that is an informative result: adjustment costs establish durability, but the strategic distortion is governed by the rival’s response and by the investing authority’s valuation of that response.

The framework now supplies a disciplined foundation for the main research question. Stronger coordination problems should be sought by varying the structural cross-effects, debt channel, mandate divergence and shock persistence, and by checking for multiple stabilising feedback equilibria. The fully forward-looking New Keynesian version is the natural next step.

A Reproduction and numerical checks

All calculations are in Julia. From the parent `australian_four_equation_game` directory, run

```
julia --project=. dynamic_two_stage/run_dynamic_game.jl
julia --project=. dynamic_two_stage/build_tex.jl
```

The first command recreates all CSV files, figures and numerical L^AT_EX macros. The second compiles this paper with the artifact-pinned Tectonic binary. The implementation checks the invertibility of every stage response system, records open-loop condition numbers, verifies closed-loop stability and runs the stationary fixed-point calculation from six initial value scales.

References

- Beetsma, R. M. W. J. and A. L. Bovenberg (1997). Central bank independence and public debt policy. *Journal of Economic Dynamics and Control* 21, 873–894. [https://doi.org/10.1016/S0165-1889\(97\)00003-1](https://doi.org/10.1016/S0165-1889(97)00003-1).
- Bulow, J. I., J. D. Geanakoplos and P. D. Klemperer (1985). Multimarket oligopoly: strategic substitutes and complements. *Journal of Political Economy* 93, 488–511. <https://doi.org/10.1086/261312>.
- Dixit, A. (1980). The role of investment in entry-deterrence. *Economic Journal* 90, 95–106. <https://doi.org/10.2307/2231658>.
- Fudenberg, D. and J. Tirole (1983). Capital as a commitment: strategic investment to deter mobility. *Journal of Economic Theory* 31, 227–250. [https://doi.org/10.1016/0022-0531\(83\)90075-3](https://doi.org/10.1016/0022-0531(83)90075-3).
- Fudenberg, D. and J. Tirole (1984). The fat-cat effect, the puppy-dog ploy, and the lean and hungry look. *American Economic Review Papers and Proceedings* 74, 361–366.
- Gnocchi, S. (2013). Monetary commitment and fiscal discretion: the optimal policy mix. *American Economic Journal: Macroeconomics* 5, 187–216. <https://doi.org/10.1257/mac.5.2.187>.

- McKibbin, W. J. (1987). Numerical solution of rational expectations models with and without strategic behaviour. Reserve Bank of Australia Research Discussion Paper 8706. <https://www.rba.gov.au/publications/rdp/1987/8706/>.
- Oudiz, G. and J. D. Sachs (1984). International policy coordination in dynamic macroeconomic models. NBER Working Paper 1417. <https://doi.org/10.3386/w1417>.
- Reserve Bank of Australia (2026). Statistical tables. <https://www.rba.gov.au/statistics/tables/>.
- Tabellini, G. (1986). Money, debt and deficits in a dynamic game. *Journal of Economic Dynamics and Control* 10, 427–442. [https://doi.org/10.1016/S0165-1889\(86\)80001-X](https://doi.org/10.1016/S0165-1889(86)80001-X).